PRODUCT USER MANUAL

For Global Ocean Wave Analysis and Forecasting Product

GLOBAL_ANALYSISFORECAST_WAV_001_027

Issue: 1.5

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2018/01/18	all	Initial version	E. Fernandez, L. Aouf	L. Nouel
1.1	2020/11/16		Extension to 10 days Use of SWIM-nadir SWH in the assimilation system	A. Dalphinet, L.Aouf, S. Law-Chune	C. Derval
1.2	2022/03/31		Update of the update frequency: twice-daily	M. Tressol, C. Derval	C. Derval
1.3	2022/11/29		New template	A. Biardeau	Copernicus Marine Product Management
1.4	2023/06/16	all	New variable + File format section updated	A. Dalphinet, M. Tressol	Copernicus Marine Product Management
1.5	2024/05/31		New variable	L. Aouf	O. Le Galloudec

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
TAC	Thematic Assembly Center
ECMWF	European Centre for Medium Range Weather forecast
GLO	Global
MFWAM	Meteo France WAve Model
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus MMarine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store
FORECAST	An assessment of the future state of the ocean variables made using numerical models.
HINCAST	An assessment of the past state of the ocean variables made using numerical models with or without data assimilation. Simulations and analyses are hindcasts.
SIMULATION:	An hindcast done without data assimilation.
NRT-ANALYSIS:	An hindcast done with data assimilation for a period very close to present. The quality and the number of available observations it's not optimal (less observations and/or less accurate than in a best-analysis).
BEST-ANALYSIS:	An hindcast done with data assimilation for a period close to present. The quality and the number of available observations it's the best available (more observations and/or more accurate than in a NRT-analysis).

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

1) INTRODUCTION

a) Summary

This document is the user manual for the Copernicus Marine global analysis and forecast product GLOBAL_ANALYSISFORECAST_WAV_001_027. It provides aggregated analyses updated twice daily with 10-day forecast with assimilation of data from the instrument SWIM-nadir of CFOSAT mission. A 2-years rolling archive is available on the Marine Data Store.

An archive of forecasts is available, see Help Center Article .

The wave products are the integrated parameters computed from the total wave spectrum (significant wave height, period, direction, Stokes drift, maximum wave height or crest...etc), as well as the following partitions: the wind wave, the primary swell wave and the secondary swell wave.

This product is global (see II.3). It is defined on a standard grid at 1/12 degree (approx. 8km).

The product is organised in 2 datasets:

- cmems_mod_glo_wav_anfc_0.083deg_PT3H-i containing 3-hourly instantaneous values for all the variables
- cmems_mod_wav_anfc_0.083deg_static -statics containing the bathymetry

It is updated as follows every day:

- the time series (analysis from the previous day and 10 days forecast) is updated at the end of the operational production process

- old obsolete forecasts are deleted from the server

The product is published on the MDS after automatic and human quality controls.
The analysis and forecasting system is described in the Quality Information Document (QUID – see in REFERENCES)

Files downloaded are in NetCDF format and follow CF-1.6 convention.

b) History of changes

Date	Description of changes and impacted product
2020/12	Temporal extension to 10 days and use of SWIM-nadir SWH in the assimilation system.
2021/12	Twice-daily delivery
2022/11	New nomenclature – new template
2023/11	Assess GLO12V4 current data compared to GLO12V3 (Upstream data change) Delivery of new crest-to-trough height variable (Updated File Format)
2024/11	Delivery of height of the highest crest (Updated File Format)

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

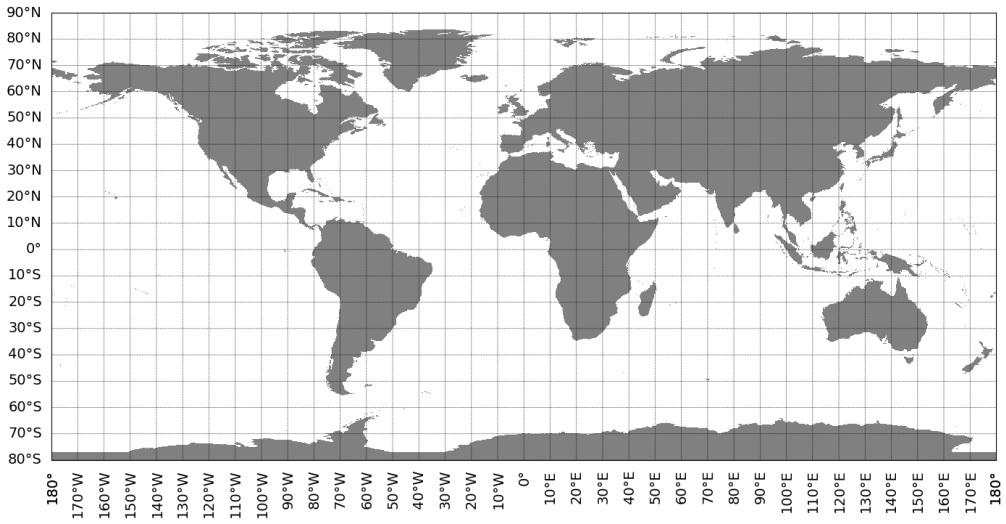
Product Lines	GLOBAL_ANALYSISFORECAST_PHY_001_027
Geographical coverage	Global : 180°W-180°E ; 89°S – 90°N
Variables	See full list of variables in this table: "Details of the datasets"
Product Type	Near Real Time - Analysis and Forecast
Available time series	-2Y to 10 Days- forecast
Temporal resolution	3-Hourly instantaneous
Update Frequency	Twice-daily
Target delivery time	Twice-daily : 00:00 UTC ;12:00 UTC
Horizontal resolution	1/12 degree
Number of vertical levels	Surface
Format	NetCDF4 CF1.6

b) Production System Description

The operational global ocean analysis and forecast system of Météo-France with a resolution of 1/12 degree is providing daily analysis and 10 days forecasts for the global ocean sea surface waves. The time series starts two years ago and is aggregated in time. This product includes 3-hourly instantaneous fields of integrated wave parameters from the total spectrum (significant height, period, direction, Stokes drift, maximum wave height and crest...etc), as well as the following partitions: the wind wave, the primary and secondary swell waves.

The analysis and forecasting system is described in the Quality Information Document (QUID – see in REFERENCES)

GLOBAL (180°W-180°E; 89°S – 90°N): 1/12°, regular grid, 4320 x 2041
This product is global with dedicated projection and spatial resolution. It is defined on a standard collocated grid at 1/12 degree (approx. 8 km). The wave parameters are interpolated from the native grid model, which is irregular with increasing distance in the latitudinal direction close to the poles.



Vertical grid	Surface
System Name	GLO12v3
Model version	MFWAM (ECWAM-IFS-41R1 source code)
Horizontal Resolution	1/12°
Vertical Resolution	Surface
Spectral resolution	24 directions and 30 frequencies
Interpolation method	
Coupling	

Bathymetry	ETOPO2/NOAA
Initial conditions	
Atmospheric forcings	Product :ECMWF wind and ice fraction Frequency: 6-hourly analysis and 3-hourly forecasts Resolution : 1/10°
Tidal constituents	
Rivers	
Data Assimilation scheme	Optimal Interpolation
Assimilated observations	Significant wave height (SWH) from altimeters Jason 3, Saral, Cryosat-2, Sentinel-3 A/B, CFOSAT and Sentinel 6 and Sentinel-1-SAR spectrum assimilation cycle : daily
Forced by another model	

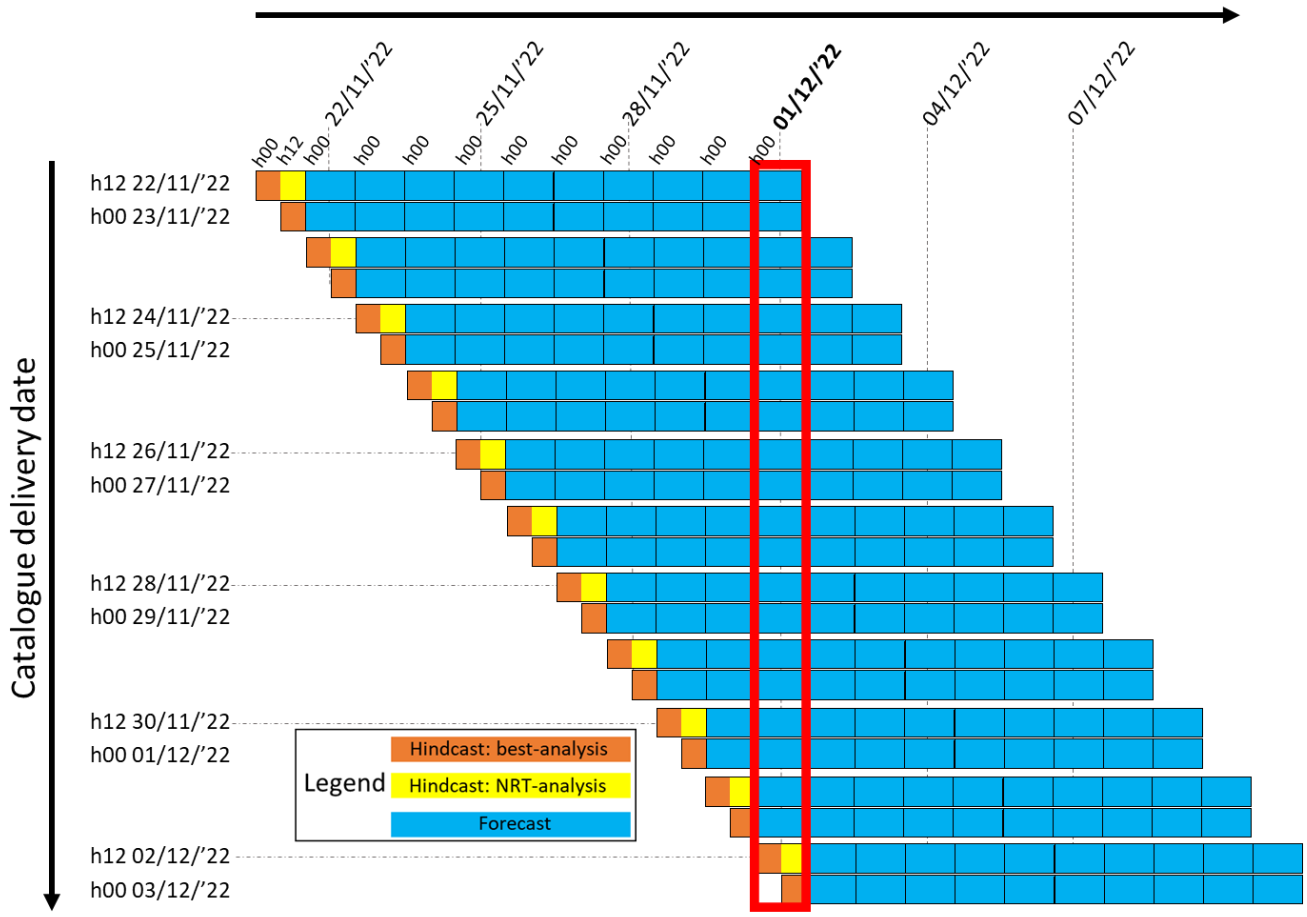
c) Details of the datasets

GLOBAL_ANALYSISFORECAST_WAV_001_027
Dataset : cmems_mod_glo_wav_anfc_0.083deg_PT3H-i contains all the variables
VHM0 [m] Spectral significant wave height (Hm0) sea_surface_wave_significant_height
VTM10 [s] Spectral moments (-1,0) wave period (Tm-10) sea_surface_wave_mean_period_from_variance_spectral_density_inverse_frequency_moment
VTM02 [s] Spectral moments (0,2) wave period (Tm02) sea_surface_wave_mean_period_from_variance_spectral_density_second_frequency_moment
VTPK [s] Wave period at spectral peak / peak period (Tp) sea_surface_wave_period_at_variance_spectral_density_maximum
VMDR [degree] Mean wave direction from (Mdir) sea_surface_wave_from_direction
VPED [degree] Wave principal direction at spectral peak sea_surface_wave_from_direction_at_variance_spectral_density_maximum
VCMX [m] Maximum crest trough wave height (Hc,max) sea_surface_wave_maximum_height
VMXL [m] Height of the highest crest sea_surface_wave_maximum_crest_height
VSDX [m s-1]

Stokes drift U sea_surface_wave_stokes_drift_x_velocity
VSDY [m s-1] Stokes drift V sea_surface_wave_stokes_drift_y_velocity
VHM0_WW [m] Spectral significant wind wave height sea_surface_wind_wave_significant_height
VTM01_WW [s] Spectral moments (0,1) wind wave period sea_surface_wind_wave_mean_period
VMDR_WW [degree] Mean wind wave direction from sea_surface_wind_wave_from_direction
VHM0_SW1 [m] Spectral significant primary swell wave height sea_surface_primary_swell_wave_significant_height
VTM01_SW1 [s] Spectral moments (0,1) primary swell wave period sea_surface_primary_swell_wave_mean_period
VMDR_SW1 [degree] Mean primary swell wave direction from sea_surface_primary_swell_wave_from_direction
VHM0_SW2 [m] Spectral significant secondary swell wave height sea_surface_secondary_swell_wave_significant_height
VTM01_SW2 [s] Spectral moments (0,1) secondary swell wave period sea_surface_secondary_swell_wave_mean_period
VMDR_SW2 [degree] Mean secondary swell wave direction from sea_surface_secondary_swell_wave_from_direction
Dataset: cmems_mod_wav_anfc_0.083deg_static contains the bathymetry.
deptho [m] Bathymetry sea_floor_depth_below_geoid

d) Catalogue updating

Every 12-hour a new 10-day forecast and 24-hr (run 00) or 12-hr (run 12) hindcast are released.
The figure shows the timeline for the delivery to the catalogue versus the validity time of the products and how the different releases for the same validity date overwrites and concatenated for building up the historical hindcast timeseries.
The red-rectangular highlights all the subsequent releases of information for the day of the 1st December 2022.



Analyses are performed 4 times a day (00, 06, 12 and 18 UTC) with updated atmospheric forcings from ECMWF.
The wave products are updated twice-daily with:

- analysis from the production day for J-24h to J
- forecast for J to J+240h

The product is updated twice- daily at 00:00 UTC &12:00 UTC

e) Time averaging

The fields are 3-hourly instantaneous at 00, 03, 06, 09, 12, 15, 18, 21 UTC

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this articles " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center"

a) Nomenclature of files when downloaded through the Subset Service

The scheme is: **datasetname_nnnnnnnnnnnnn.nc**

where:

- **datasetname**: as described previously
- **nnnnnnnnnnnnn**: 13-digit integer corresponding to the current time (download time) in milliseconds since January 1, 1970 midnight UTC.
- **.nc**: standard NetCDF filename extension.

Example: global-analysis-forecast-wav-001-027_1669311372719.nc

b) Nomenclature of original files

The scheme is: modelname_${date1}-R${date2}H.ncmodelname_${date1}00-R${date2}.nc
where:

- **modelname**: mfwamglocep
- **date1**: integer (format YYYYMMDDHH) corresponding to the field date
- **date2**: integer (format YYYYMMDD_HH) corresponding to the analysis date

date1: integer (format YYYYMMDD) corresponding to the field date

- **date2**: integer (format YYYYMMDD) corresponding to the analysis date
- **.nc**: standard NetCDF filename extension.

Example:
- mfwamglocep_2023020600_R20230207_00H.nc : 6th June 2023 fields from 3H to 12H of bulletin 20230207 - network 00H
- mfwamglocep_2023020612_R20230207_00H.nc : 6th June 2023 fields from 15H to 24H of bulletin 20230207 - network 00H
- mfwamglocep_2023020612_R20230207_12H.nc : 6th June 2023 fields from 15H to 24H of bulletin 20230207 - network 12H (replace the previous one when network available)

- mfwamglocep_2023020700_R20230207_12H.nc : 7th June 2023 fields from 3H to 12H of bulletin 20230207 - network 12HExample: mfwamglocep_2021050700_R20210508.nc (7 May 2021 field from 8 May 2021 bulletin)

c) Additional Information

Real_Value = (Display_Value X scale_factor) + add_offset
The missing value for this product is: -32767s
Land mask are equal to "_FillValue" (see variable attribute on NetCDF file).

5) FILE SIZE

DATASET NAME	FILE NAME	SIZE [GB]
cmems_mod_glo_wav_anfc_0.083deg_PT3H-i	mfwamglocep_${date1}00_R${date2}.nc	0.6
cmems_mod_wav_anfc_0.083deg_static	GLO-MFC_001_027_bathy.nc	0.07

Where :

- ${date1} is the field date : format YYYYMMDDHH,
- ${date2} is the bulletin date : format (YYYYMMDD_HH),${date1} is the date : format YYYYMMDD,

6) STRUCTURE OF FILES

Example of the header of output NetCDF file is inserted in annex

7) REFERENCES

Quality Information Document (QUID) : CMEMS-GLO-QUID-001-027.pdf (copernicus.eu)

8) ANNEX

```
netcdf mfwamglocep_2023101012_R20231002_00H.nc {
  dimensions:
    time = UNLIMITED;   // (4 currently)
    longitude = 4320;
    latitude = 2041;
  variables:
    int time(time=4);
      :standard_name = "time";
      :long_name = "time";
      :axis = "T";
      :step = 3; // int
      :units = "hours since 1950-01-01";
      :calendar = "gregorian";
      :_ChunkSizes = 1024U; // uint

    double longitude(longitude=4320);
      :standard_name = "longitude";
      :long_name = "longitude coordinate";
      :units = "degrees_east";
      :axis = "X";
      :step = 0.08332825f; // float

    double latitude(latitude=2041);
      :standard_name = "latitude";
      :long_name = "latitude coordinate";
      :units = "degrees_north";
      :axis = "Y";
      :step = 0.08333588f; // float

    short VHM0(time=4, latitude=2041, longitude=4320);
      :_FillValue = -32767S; // short
      :standard_name = "sea_surface_wave_significant_height";
      :long_name = "Spectral significant wave height (Hm0)";
      :units = "m";
      :cell_methods = "time:point area:mean";
      :type_of_analysis = "spectral analysis";
      :WMO_code = 100; // int
      :add_offset = 0.0f; // float
      :scale_factor = 0.01f; // float
      :missing_value = -32767S; // short
      :_ChunkSizes = 1U, 1021U, 2160U; // uint

    short VHM0_WW(time=4, latitude=2041, longitude=4320);
      :_FillValue = -32767S; // short
      :standard_name = "sea_surface_wind_wave_significant_height";
      :long_name = "Spectral significant wind wave height";
      :units = "m";
      :cell_methods = "time:point area:mean";
      :type_of_analysis = "spectral analysis";
      :WMO_code = 102; // int
      :add_offset = 0.0f; // float
```

```
    :scale_factor = 0.01f; // float
    :missing_value = -32767S; // short
    :_ChunkSizes = 1U, 1021U, 2160U; // uint

  short VTM01_SW1(time=4, latitude=2041, longitude=4320);
    :_FillValue = -32767S; // short
    :standard_name = "sea_surface_primary_swell_wave_mean_period";
    :long_name = "Spectral moments (0,1) primary swell wave period";
    :units = "s";
    :cell_methods = "time:point area:mean";
    :type_of_analysis = "spectral analysis";
    :WMO_code = 226; // int
    :add_offset = 0.0f; // float
    :scale_factor = 0.01f; // float
    :missing_value = -32767S; // short
    :_ChunkSizes = 1U, 1021U, 2160U; // uint

  short VTM01_SW2(time=4, latitude=2041, longitude=4320);
    :_FillValue = -32767S; // short
    :standard_name = "sea_surface_secondary_swell_wave_mean_period";
    :long_name = "Spectral moments (0,1) secondary swell wave period";
    :units = "s";
    :cell_methods = "time:point area:mean";
    :type_of_analysis = "spectral analysis";
    :WMO_code = 227; // int
    :add_offset = 0.0f; // float
    :scale_factor = 0.01f; // float
    :missing_value = -32767S; // short
    :_ChunkSizes = 1U, 1021U, 2160U; // uint

  short VTM10(time=4, latitude=2041, longitude=4320);
    :_FillValue = -32767S; // short
    :standard_name =
"sea_surface_wave_mean_period_from_variance_spectral_density_inverse_frequency_mome
nt";
    :long_name = "Spectral moments (-1,0) wave period (Tm-10)";
    :units = "s";
    :cell_methods = "time:point area:mean";
    :type_of_analysis = "spectral analysis";
    :WMO_code = 201; // int
    :add_offset = 0.0f; // float
    :scale_factor = 0.01f; // float
    :missing_value = -32767S; // short
    :_ChunkSizes = 1U, 1021U, 2160U; // uint

  short VHM0_SW1(time=4, latitude=2041, longitude=4320);
    :_FillValue = -32767S; // short
    :standard_name = "sea_surface_primary_swell_wave_significant_height";
    :long_name = "Spectral significant primary swell wave height";
    :units = "m";
    :cell_methods = "time:point area:mean";
    :type_of_analysis = "spectral analysis";
    :WMO_code = 202; // int
    :add_offset = 0.0f; // float
```

```
  :scale_factor = 0.01f; // float
  :missing_value = -32767S; // short
  :_ChunkSizes = 1U, 1021U, 2160U; // uint

short VHM0_SW2(time=4, latitude=2041, longitude=4320);
  :_FillValue = -32767S; // short
  :standard_name = "sea_surface_secondary_swell_wave_significant_height";
  :long_name = "Spectral significant secondary swell wave height";
  :units = "m";
  :cell_methods = "time:point area:mean";
  :type_of_analysis = "spectral analysis";
  :WMO_code = 203; // int
  :add_offset = 0.0f; // float
  :scale_factor = 0.01f; // float
  :missing_value = -32767S; // short
  :_ChunkSizes = 1U, 1021U, 2160U; // uint

short VTPK(time=4, latitude=2041, longitude=4320);
  :_FillValue = -32767S; // short
  :standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum";
  :long_name = "Wave period at spectral peak / peak period (Tp)";
  :units = "s";
  :cell_methods = "time:point area:mean";
  :type_of_analysis = "spectral analysis";
  :WMO_code = 204; // int
  :add_offset = 0.0f; // float
  :scale_factor = 0.01f; // float
  :missing_value = -32767S; // short
  :_ChunkSizes = 1U, 1021U, 2160U; // uint

short VSDX(time=4, latitude=2041, longitude=4320);
  :_FillValue = -32767S; // short
  :standard_name = "sea_surface_wave_stokes_drift_x_velocity";
  :long_name = "Stokes drift U";
  :units = "m s-1";
  :cell_methods = "time:point area:mean";
  :type_of_analysis = "spectral analysis";
  :WMO_code = 215; // int
  :add_offset = 0.0f; // float
  :scale_factor = 0.01f; // float
  :missing_value = -32767S; // short
  :_ChunkSizes = 1U, 1021U, 2160U; // uint

short VSDY(time=4, latitude=2041, longitude=4320);
  :_FillValue = -32767S; // short
  :standard_name = "sea_surface_wave_stokes_drift_y_velocity";
  :long_name = "Stokes drift V";
  :units = "m s-1";
  :cell_methods = "time:point area:mean";
  :type_of_analysis = "spectral analysis";
  :WMO_code = 216; // int
  :add_offset = 0.0f; // float
  :scale_factor = 0.01f; // float
  :missing_value = -32767S; // short
```

```
    :_ChunkSizes = 1U, 1021U, 2160U; // uint

  short VCMX(time=4, latitude=2041, longitude=4320);
    :_FillValue = -32767S; // short
    :standard_name = "sea_surface_wave_maximum_height";
    :long_name = "Maximum crest trough wave height";
    :units = "m";
    :cell_methods = "time:point area:mean";
    :type_of_analysis = "spectral analysis";
    :WMO_code = 218; // int
    :add_offset = 0.0f; // float
    :scale_factor = 0.01f; // float
    :missing_value = -32767S; // short
    :_ChunkSizes = 1U, 1021U, 2160U; // uint

     short VMXL(time, latitude, longitude) ;
            VMXL:long_name = "Height of the highest crest" ;
            VMXL:units = "m" ;
            VMXL:add_offset = 0.f ;
            VMXL:scale_factor = 0.01f ;
            VMXL:_FillValue = -32767s ;
            VMXL:standard_name = "sea_surface_wave_maximum_crest_height" ;
            VMXL:cell_methods = "time:point area:mean" ;
            VMXL:type_of_analysis = "spectral analysis" ;

  short VTM02(time=4, latitude=2041, longitude=4320);
    :_FillValue = -32767S; // short
    :standard_name =
"sea_surface_wave_mean_period_from_variance_spectral_density_second_frequency_mome
nt";
    :long_name = "Spectral moments (0,2) wave period (Tm02)";
    :units = "s";
    :cell_methods = "time:point area:mean";
    :type_of_analysis = "spectral analysis";
    :WMO_code = 221; // int
    :add_offset = 0.0f; // float
    :scale_factor = 0.01f; // float
    :missing_value = -32767S; // short
    :_ChunkSizes = 1U, 1021U, 2160U; // uint

  short VTM01_WW(time=4, latitude=2041, longitude=4320);
    :_FillValue = -32767S; // short
    :standard_name = "sea_surface_wind_wave_mean_period";
    :long_name = "Spectral moments (0,1) wind wave period";
    :units = "s";
    :cell_methods = "time:point area:mean";
    :type_of_analysis = "spectral analysis";
    :WMO_code = 223; // int
    :add_offset = 0.0f; // float
    :scale_factor = 0.01f; // float
    :missing_value = -32767S; // short
    :_ChunkSizes = 1U, 1021U, 2160U; // uint

  short VMDR_WW(time=4, latitude=2041, longitude=4320);
```

```
  :_FillValue = -32767S; // short
  :standard_name = "sea_surface_wind_wave_from_direction";
  :long_name = "Mean wind wave direction from";
  :units = "degree";
  :cell_methods = "time:point area:mean";
  :type_of_analysis = "spectral analysis";
  :WMO_code = 101; // int
  :add_offset = 180.0f; // float
  :scale_factor = 0.01f; // float
  :missing_value = -32767S; // short
  :_ChunkSizes = 1U, 1021U, 2160U; // uint

short VMDR_SW1(time=4, latitude=2041, longitude=4320);
  :_FillValue = -32767S; // short
  :standard_name = "sea_surface_primary_swell_wave_from_direction";
  :long_name = "Mean primary swell wave direction from";
  :units = "degree";
  :cell_methods = "time:point area:mean";
  :type_of_analysis = "spectral analysis";
  :WMO_code = 107; // int
  :add_offset = 180.0f; // float
  :scale_factor = 0.01f; // float
  :missing_value = -32767S; // short
  :_ChunkSizes = 1U, 1021U, 2160U; // uint

short VMDR_SW2(time=4, latitude=2041, longitude=4320);
  :_FillValue = -32767S; // short
  :standard_name = "sea_surface_secondary_swell_wave_from_direction";
  :long_name = "Mean secondary swell wave direction from";
  :units = "degree";
  :cell_methods = "time:point area:mean";
  :type_of_analysis = "spectral analysis";
  :WMO_code = 109; // int
  :add_offset = 180.0f; // float
  :scale_factor = 0.01f; // float
  :missing_value = -32767S; // short
  :_ChunkSizes = 1U, 1021U, 2160U; // uint

short VMDR(time=4, latitude=2041, longitude=4320);
  :_FillValue = -32767S; // short
  :standard_name = "sea_surface_wave_from_direction";
  :long_name = "Mean wave direction from (Mdir)";
  :units = "degree";
  :cell_methods = "time:point area:mean";
  :type_of_analysis = "spectral analysis";
  :WMO_code = 200; // int
  :add_offset = 180.0f; // float
  :scale_factor = 0.01f; // float
  :missing_value = -32767S; // short
  :_ChunkSizes = 1U, 1021U, 2160U; // uint

short VPED(time=4, latitude=2041, longitude=4320);
  :_FillValue = -32767S; // short
```

```
      :standard_name =
"sea_surface_wave_from_direction_at_variance_spectral_density_maximum";
      :long_name = "Wave principal direction at spectral peak";
      :units = "degree";
      :cell_methods = "time:point area:mean";
      :type_of_analysis = "spectral analysis";
      :add_offset = 180.0f; // float
      :scale_factor = 0.01f; // float
      :missing_value = -32767S; // short
      :_ChunkSizes = 1U, 1021U, 2160U; // uint

  // global attributes:
  :Conventions = "CF-1.6";
  :product = "GLOBAL_ANALYSIS_FORECAST_WAV_001_027";
  :institution = "METEO-FRANCE";
  :references = "http://marine.copernicus.eu";
  :credit = "E.U. Copernicus Marine Service Information (CMEMS)";
  :contact = "servicedesk.cmems@mercator-ocean.eu";
  :producer = "CMEMS - Global Monitoring and Forecasting Centre";
  :area = "GLO";
  :geospatial_lon_min = -180.0f; // float
  :geospatial_lon_max = 179.9167f; // float
  :geospatial_lon_step = 0.08332825f; // float
  :geospatial_lon_units = "degree";
  :geospatial_lat_min = -80.0f; // float
  :geospatial_lat_max = 90.0f; // float
  :geospatial_lat_step = 0.08333588f; // float
  :geospatial_lat_units = "degree";
}
```